

HOBART, OK, USA

WMO: 723525

Lat: 34.989N

Lon: 99.053W

Elev: 474

StdP: 95.76

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93986

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.7	-7.3	-16.3	1.0	-6.5	-14.0	1.2	-3.6	14.9	7.7	13.6	8.0	5.1	20	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	39.7	22.3	38.2	22.3	36.8	22.3	25.0	33.0	24.2	32.8	23.7	32.3	6.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.6	27.9	22.2	17.9	27.2	21.5	17.1	26.7	78.4	33.0	75.6	32.6	73.3	32.2	31.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.0	11.8	10.8	-14.0	41.7	2.5	1.8	-15.8	43.0	-17.2	44.1	-18.6	45.1	-20.4	46.4
			-14.4	26.5	2.5	1.9	-16.1	27.9	-17.6	29.0	-19.0	30.0	-20.8	31.4
				DB										
				WB										

Monthly Climatic Design Conditions

			Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
			(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	4.1	6.1	10.8	15.4	21.2	26.6	29.0	28.4	24.0	17.1	10.3	4.6
(7)		DBStd	10.08	5.28	6.03	5.72	4.78	4.62	3.25	2.97	3.09	4.15	4.98	5.26	5.14
(8)		HDD10.0	642	193	134	59	11	1	0	0	0	7	60	177	
(9)		HDD18.3	1919	442	342	239	110	25	1	0	7	83	245	424	
(10)		CDD10.0	3020	9	26	84	174	347	497	588	569	421	226	68	11
(11)		CDD18.3	1255	0	1	6	23	113	248	329	311	177	44	3	0
(12)		CDH23.3	15264	2	17	68	272	1240	2968	4407	3943	1870	436	39	2
(13)		CDH26.7	8259	0	3	13	78	557	1607	2631	2298	922	144	7	0
(14)	Wind	WSAvg	5.6	5.5	5.8	6.1	6.1	5.5	5.8	5.2	4.8	5.1	5.7	5.7	5.3
(15)	Precipitation	PrecAvg	674	19	26	44	54	112	85	58	65	83	69	38	27
(16)		PrecMax	920	76	76	130	177	306	247	269	191	278	265	170	112
(17)		PrecMin	337	0	0	0	0	3	14	1	0	0	0	0	0
(18)		PrecStd	151	20	22	31	39	81	60	49	50	70	57	37	29
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.6	25.7	28.5	32.1	37.3	40.6	41.6	41.4	39.1	32.9	27.2	21.3
(20)			MCWB	11.9	13.3	15.4	17.7	20.8	21.8	22.2	21.8	22.5	19.9	16.6	12.1
(21)		2%	DB	18.8	22.1	25.1	28.9	34.6	37.7	39.7	39.4	36.2	30.4	23.9	17.9
(22)			MCWB	9.9	12.7	15.0	17.8	20.8	22.0	22.2	22.5	21.7	19.2	15.3	11.8
(23)		5%	DB	16.0	18.7	22.7	26.5	32.3	36.0	38.3	37.9	34.1	27.9	21.2	15.5
(24)			MCWB	8.8	11.5	15.0	17.0	20.4	22.3	22.5	22.4	21.5	18.4	14.1	9.7
(25)		10%	DB	13.1	16.0	20.3	24.2	29.7	34.2	36.8	36.2	32.0	25.7	18.7	12.9
(26)			MCWB	7.4	10.3	13.9	16.6	20.0	22.4	22.6	22.4	20.9	17.2	13.3	8.4
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.1	16.7	19.7	22.0	24.0	26.0	25.9	25.3	25.2	22.8	19.3	16.1
(28)			MCDWB	18.5	20.8	24.3	26.5	31.3	33.4	33.5	33.1	33.9	28.6	22.9	18.1
(29)		2%	WB	11.6	14.6	18.0	20.4	22.9	24.7	24.8	24.5	23.7	21.4	17.0	13.2
(30)			MCDWB	16.7	19.2	22.2	25.0	30.1	32.7	33.2	33.1	31.2	26.4	21.3	16.1
(31)		5%	WB	9.5	12.7	16.6	19.2	22.1	24.0	24.1	23.9	22.9	20.2	15.6	10.6
(32)			MCDWB	15.1	17.8	20.4	23.8	28.9	32.3	33.0	32.7	30.1	24.8	19.6	14.4
(33)		10%	WB	7.6	10.5	14.8	17.9	21.3	23.3	23.6	23.4	22.1	19.0	14.1	8.3
(34)			MCDWB	12.9	15.6	19.4	22.3	27.8	31.3	32.5	32.1	28.9	23.9	17.9	12.5
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.9	13.0	13.3	13.6	13.2	13.0	13.4	13.1	12.7	12.9	12.6	11.7
(36)			MCDBR	18.0	17.8	16.5	16.9	16.1	15.2	15.4	15.4	15.0	15.4	15.8	15.8
(37)			MCWBR	10.9	10.5	9.3	8.6	5.9	4.7	3.8	4.4	5.1	7.0	8.8	9.9
(38)		5% WB	MCDBR	16.3	15.4	13.1	13.0	13.6	12.9	13.1	12.9	12.2	11.7	13.0	13.5
(39)			MCWBR	10.7	10.4	8.4	8.3	6.4	5.1	4.3	4.3	4.9	6.6	8.3	9.6
(40)	Clear-Sky Solar Irradiance	taub	0.282	0.294	0.329	0.363	0.388	0.413	0.413	0.407	0.374	0.344	0.308	0.285	
(41)		taud	2.529	2.503	2.421	2.354	2.356	2.324	2.355	2.319	2.388	2.459	2.494	2.519	
(42)		Ebn at Noon	935	958	943	921	897	871	868	868	884	886	891	908	
(43)		Edh at Noon	80	92	109	123	124	128	124	126	112	95	83	76	
(44)	All-Sky Solar Radiation	RadAvg	2.99	3.61	4.79	5.87	6.34	7.01	7.00	6.35	5.31	4.04	3.12	2.54	
(45)		RadStd	0.29	0.48	0.31	0.45	0.66	0.59	0.39	0.37	0.36	0.54	0.39	0.26	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page